

BIG DATA PROJECT 2

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docker

STEPS

- ▶ STEP 1: Generated two files through SUMO - **emission_data.xml** and **fcd_data.xml**. File **osm.net.sml** contains the data required for converting x and y values into lat and lon values for the map.
- ▶ STEP 2: written **docker-compose** file. Run:
docker network create bde
docker compose up -d

☐	↓	Name	Container ID	Image	Port(s)	CPU (%)	Memory	Actions
☐	●	flink-taskmar	d32624fce8b0	flink:latest		1.86%	1.04GB /	☐ ⋮
☐	●	flink-jobman	da48063a3b79	flink:latest	8081:8081 ↗	1.85%	1.03GB /	☐ ⋮
☐	●	spark-worker	80e32b0368e1	bde2020/s	8083:8081 ↗	0.14%	205.8ME	☐ ⋮
☐	●	spark-worker	fcfcf925711e	bde2020/s	8082:8081 ↗	0.12%	175.4ME	☐ ⋮
☐	●	kafka-1	3db2e62428f4	apache/kaf	9092:29092 ↗	2.13%	1.05GB /	☐ ⋮
☐	●	kafka-2	7b9a1e9cf74b	apache/kaf	9094:29092 ↗	2.13%	1.05GB /	☐ ⋮
☐	●	spark-master	f4fef9f46ca0	bde2020/s	7077:7077 ↗ Show all ports (2)	0.12%	312.4ME	☐ ⋮

Example of the data from the **emission.xml** file:

```
<vehicle id="truck7" eclass="HBEFA4/RT_1e7.5t_Euro-VI_A-C"  
CO2="11473.89" CO="1.61" HC="0.20" NOx="5.72" PMx="3.28" fuel="3688.22"  
electricity="0.00" noise="81.89" route="!truck7!var#1" type="truck_truck"  
waiting="0.00" lane="72247241#0_0" pos="7.20" speed="24.14" angle="108.73"  
x="3297.93" y="10006.73"/>
```

Example of the data from the **fcd.xml** file:

```
<vehicle id="truck4" x="3405.01" y="8052.51" angle="248.91"  
type="truck_truck" speed="14.18" pos="7.20" lane="-151362347#1_0"  
slope="0.00"/>
```

STEPS

► STEP 3: Written files:

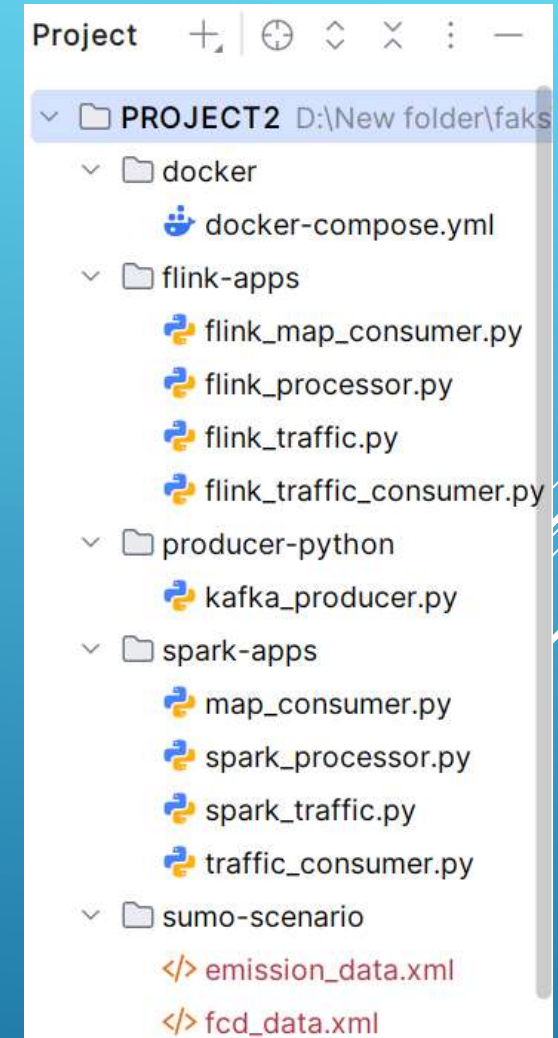
kafka_producer.py – for merging the two .xml files, converting x and y values to lat and lon values, extracting the significant values from the .xml files, sending data to the **madrid-emission** Kafka topic.

spark_processor.py and **spark_traffic.py** – for **Spark** processing data from the **madrid-emission** topic and sending it to the **madrid-results** or **madrid-traffic** topic.

flink_processor.py and **flink_traffic.py** – the **Flink** versions, processing data from the **madrid-emission** topic and sending it to the **flink-results** or **flink-traffic** topic.

map_consumer.py and **traffic_consumer.py** – for visualizing the map of Madrid with pollution or traffic circles, taking data from the **madrid-results** or **madrid-traffic** topic.

flink_map_consumer.py and **flink_traffic_consumer.py** – Flink versions for the visualization, taking data from **flink-results** or **flink-traffic** topic.



STEPS

- ▶ STEP 4: Installing python inside a flink-jobmanager and flink-taskmanager.
- ▶ STEP 5: Installing apache-flink inside a flink-jobmanager and flink-taskmanager.
- ▶ STEP 6: Downloading of Kafka connector JAR file.

```
docker exec -u root flink-jobmanager wget -P /opt/flink/lib/  
https://repo1.maven.org/maven2/org/apache/flink/flink-sql-  
connector-kafka/4.0.1-2.0/flink-sql-connector-kafka-4.0.1-2.0.jar
```

```
docker exec -u root flink-taskmanager wget -P /opt/flink/lib/  
https://repo1.maven.org/maven2/org/apache/flink/flink-sql-  
connector-kafka/4.0.1-2.0/flink-sql-connector-kafka-4.0.1-2.0.jar
```

STEPS

- STEP 7: Configurations for the scripts:

flink_map_consumer.py: --broker localhost:9092 --pollutant CO2 --output flink_pollution_map.html --refresh 30

flink_traffic_consumer.py: --broker localhost:9092 --output flink_traffic_map.html --refresh 30

map_consumer.py: --broker localhost:9092 --pollutant CO2 --output pollution_map.html --refresh

traffic_consumer.py: --broker localhost:9092 --output traffic_map.html --refresh 30

kafka_producer.py: --file "D:\New folder\faks\MASTER\BIGDATA\PROJECT2\sumo-scenario\emission_data.xml" --broker localhost:9092 --delay 0.05

!Delay could be modified for a slower or faster production!

RUNNING

- TOPIC CREATION (example for the **flink-results** topic):
`docker exec kafka-1 /opt/kafka/bin/kafka-topics.sh --create --topic flink-results --bootstrap-server kafka-1:9092 --partitions 8 --replication-factor 1`
- RUN COMMANDS:
spark_processor.py: `docker exec -it spark-master /spark/bin/spark-submit --packages org.apache.spark:spark-sql-kafka-0-10_2.12:3.1.1 /app/spark_processor.py "10 minutes" "2 minutes" "CO2" "all"`

spark_traffic.py: `docker exec -it spark-master /spark/bin/spark-submit --packages org.apache.spark:spark-sql-kafka-0-10_2.12:3.1.1 /app/spark_traffic.py "2 minutes" "30 seconds" "all"`

flink_processor.py: `docker exec flink-jobmanager flink run --python /opt/flink-apps/flink_processor.py -- 120 30 CO2 all`

flink_traffic.py: `docker exec flink-jobmanager flink run --python /opt/flink-apps/flink_traffic.py -- 120 30 all`

RUNNING

- Run order:
processor script -> producer -> map script
- Example:
spark_processor.py -> kafka_producer.py -> map_consumer.py

Example of the output from the **spark_processor.py** file:

```
26/03/08 21:28:40 INFO SharedState: Warehouse path is 'file:/spark-warehouse'.  
[Spark] Streaming started. Writing to topic: madrid-results  
[Spark] Checkpoint directory: /tmp/spark-checkpoint
```

Example of the output from the **flink_processor.py** file:

```
PS D:\New folder\faks\MASTER\BIGDATA\PROJECT2\flink-apps> docker exec flink-jobmanager flink run --python /opt/flink-apps/flink_processor.py -- 600  
120 CO2 all  
WARNING: Unknown module: jdk.compiler specified to --add-exports  
WARNING: Unknown module: jdk.compiler specified to --add-exports  
WARNING: Unknown module: jdk.compiler specified to --add-exports  
WARNING: Unknown module: jdk.compiler specified to --add-exports  
WARNING: Unknown module: jdk.compiler specified to --add-exports  
INFO:__main__: [Flink Pollution] Window type : sliding  
INFO:__main__: [Flink Pollution] Window size : 600s  
INFO:__main__: [Flink Pollution] Slide step : 120s  
INFO:__main__: [Flink Pollution] Pollutants : ['CO2']  
INFO:__main__: [Flink Pollution] Vehicle : all  
INFO:apache_beam.typehints.native_type_compatibility:Using Any for unsupported type: typing.Sequence[~T]  
INFO:apache_beam.io.gcp.bigquery:No module named google.cloud.bigquery_storage_v1. As a result, the ReadFromBigQuery transform *CANNOT* be used with  
'method=DIRECT_READ'.  
INFO:__main__: [Flink Pollution] Starting job. Writing to topic: flink-results  
Job has been submitted with JobID e889d34284728d11ddcd5de9ba97a66f
```

Example of the output from the `kafka_producer.py` file:

```
[Producer] Connected to Kafka broker: localhost:9092
[Producer] Streaming from: D:\New folder\faks\MASTER\BIGDATA\PROJECT2\sumo-scenario\emission_data.xml
[Producer] Coordinate check: (3701.62, 9001.43) → (40.42064, -3.708337)
[Producer] Expected: somewhere around (40.37-40.45, -3.75--3.58)
[Producer] Sent 1000 records | timestep=10.0s
[Producer] Sent 2000 records | timestep=14.0s
[Producer] Sent 3000 records | timestep=17.0s
[Producer] Sent 4000 records | timestep=20.0s
[Producer] Sent 5000 records | timestep=22.0s
[Producer] Sent 6000 records | timestep=24.0s
[Producer] Sent 7000 records | timestep=26.0s
[Producer] Sent 8000 records | timestep=28.0s
[Producer] Sent 9000 records | timestep=30.0s
```

End:

```
FINISHED! Total number of 4760595 messages sent on madrid-emission.
```

```
Process finished with exit code 0
```

Example of the output from the **flink_map_consumer.py** file:

```
[Flink Pollution Consumer] Connected to Kafka broker: localhost:9092
[Flink Pollution Consumer] Listening on topic: flink-results
[Flink Pollution Consumer] Visualizing pollutant: CO2
[Flink Pollution Map] Rendered 1186 grid cells | pollutant=CO2
[Flink Pollution Map] Saved → flink_pollution_map.html
[Flink Pollution Map] Rendered 1210 grid cells | pollutant=CO2
[Flink Pollution Map] Saved → flink_pollution_map.html
[Flink Pollution Map] Rendered 1215 grid cells | pollutant=CO2
[Flink Pollution Map] Saved → flink_pollution_map.html
[Flink Pollution Map] Rendered 1215 grid cells | pollutant=CO2
[Flink Pollution Map] Saved → flink_pollution_map.html
```

Example of the output from the **map_consumer.py** file:

```
[Map] Rendered 614 grid cells | pollutant=CO2
[Map] Saved → pollution_map.html
[Map] Rendered 619 grid cells | pollutant=CO2
[Map] Saved → pollution_map.html
[Map] Rendered 627 grid cells | pollutant=CO2
[Map] Saved → pollution_map.html
[Map] Rendered 634 grid cells | pollutant=CO2
[Map] Saved → pollution_map.html
[Map] Rendered 640 grid cells | pollutant=CO2
[Map] Saved → pollution_map.html
```

Example of the output from the **traffic_consumer.py** file:

```
[Traffic Consumer] Connected to Kafka broker: localhost:9092
[Traffic Consumer] Listening on topic: madrid-traffic
[Traffic Map] Rendered 94 grid cells
[Traffic Map] Saved → traffic_map.html
[Traffic Map] Rendered 130 grid cells
[Traffic Map] Saved → traffic_map.html
[Traffic Map] Rendered 160 grid cells
[Traffic Map] Saved → traffic_map.html
[Traffic Map] Rendered 190 grid cells
[Traffic Map] Saved → traffic_map.html
[Traffic Map] Rendered 214 grid cells
[Traffic Map] Saved → traffic_map.html
```

!After each run, before starting
the new one, reset the topics and
checkpoints!

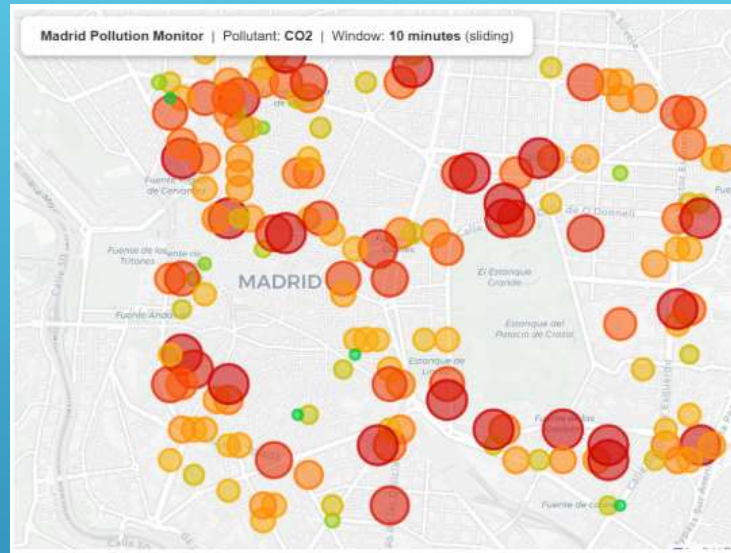
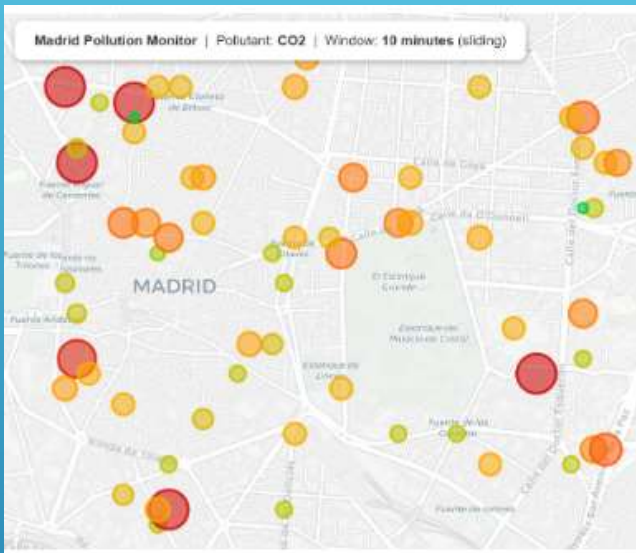
Deleting **Spark** checkpoints:

```
docker exec spark-master rm -rf /tmp/spark-checkpoint  
docker exec spark-master rm -rf /tmp/spark-traffic-checkpoint
```

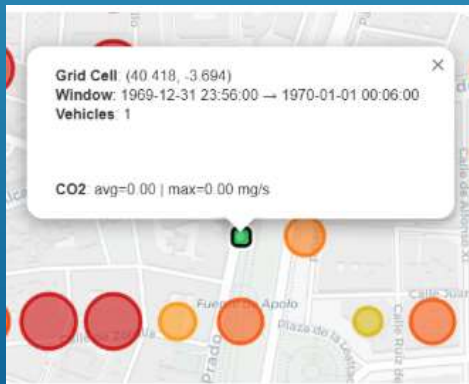
Example of restarting **Kafka** topics:

```
docker exec kafka-1 /opt/kafka/bin/kafka-topics.sh --delete --topic flink-results --bootstrap-server  
kafka-1:9092  
docker exec kafka-1 /opt/kafka/bin/kafka-topics.sh --create --topic flink-results --bootstrap-server  
kafka-1:9092 --partitions 8 --replication-factor 1  
docker exec kafka-1 /opt/kafka/bin/kafka-topics.sh --delete --topic flink-traffic --bootstrap-server  
kafka-1:9092  
docker exec kafka-1 /opt/kafka/bin/kafka-topics.sh --create --topic flink-traffic --bootstrap-server  
kafka-1:9092 --partitions 8 --replication-factor 1  
docker exec kafka-1 /opt/kafka/bin/kafka-topics.sh --delete --topic madrid-emission --bootstrap-server  
kafka-1:9092  
docker exec kafka-1 /opt/kafka/bin/kafka-topics.sh --create --topic madrid-emission --bootstrap-server  
kafka-1:9092 --partitions 8 --replication-factor 1
```

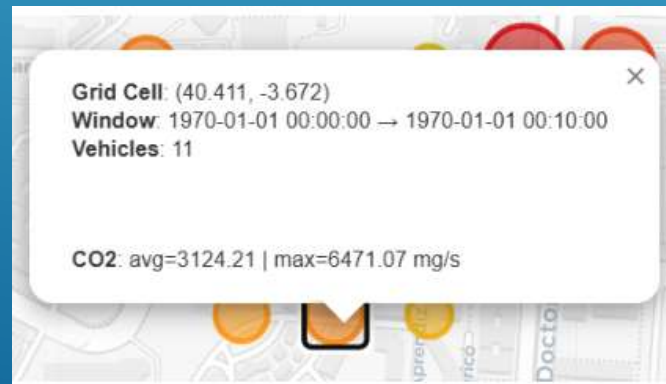

Example – Spark Pollution Job



Low pollution:



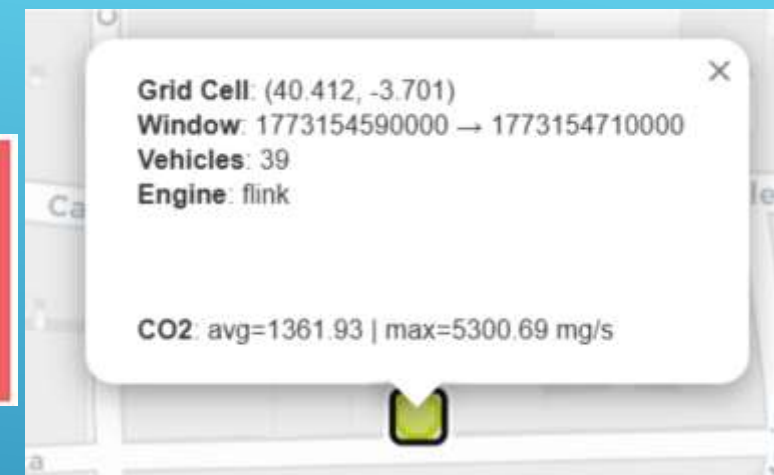
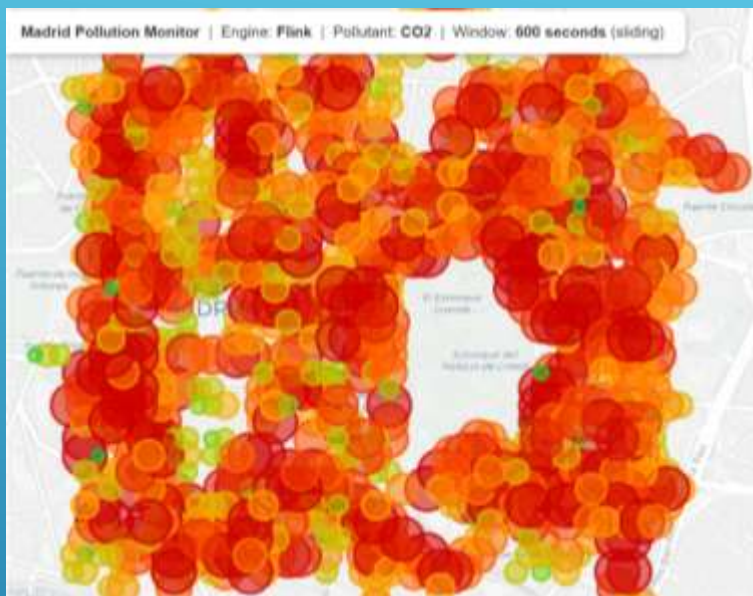
Medium pollution:



High pollution:



Example – Flink Pollution Job



Completed Job List					
Job Name	Start Time	Duration	End Time	Tasks	Status
SUMO_Madrid_Flink_Pollution	2026-03-10 15:55:46.160	4m 27s 180ms	2026-03-10 16:00:13.340	4/4	CANCELED

Status is cancelled because I cancelled it manually!

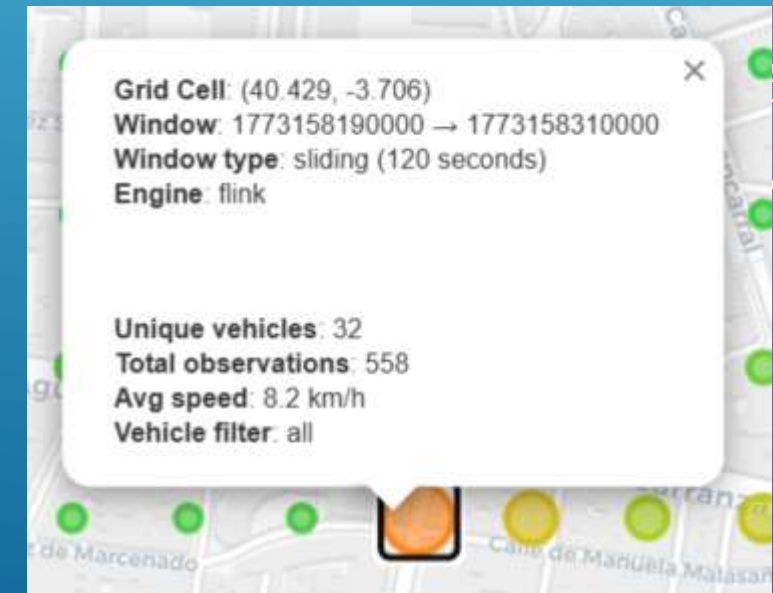
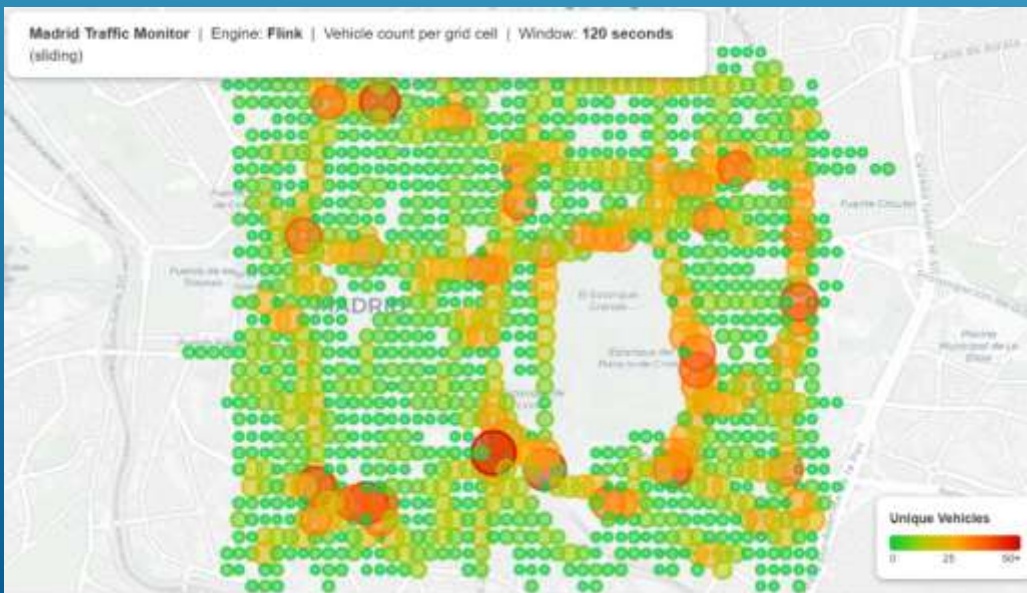
Example – Spark Traffic Job



Changed to 2 minutes window to speed up the visualization!



Example – Flink Traffic Job



Example – Flink Traffic Job

Apache Flink Dashboard

Version: 2.2.0 Commit: 5a33609 @ 2025-11-27T02:56:12+01:00 Message: 0

Overview

Jobs

- Running Jobs
- Completed Jobs

Task Managers

Job Manager

Submit New Job

Available Task Slots

0

Total Task Slots: 2 Task Managers: 1

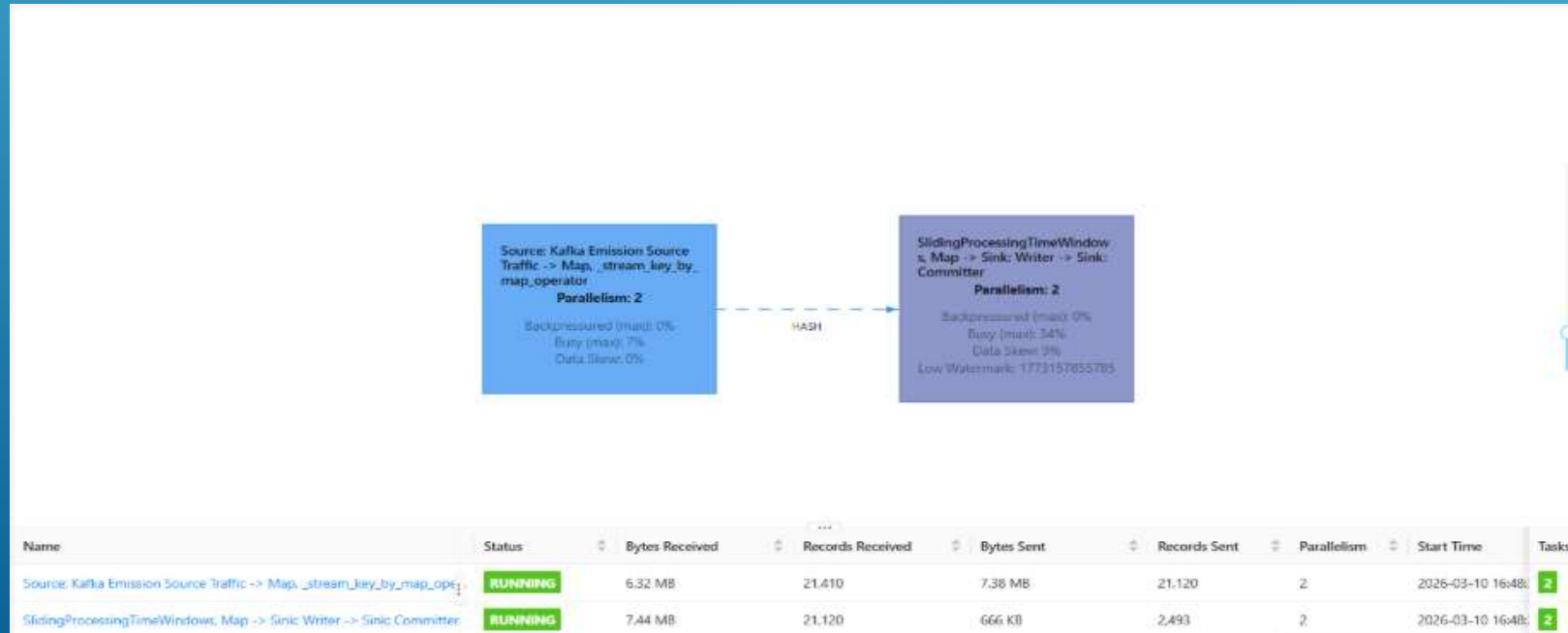
Running Jobs

1

Finished: 0 Canceled: 3 Failed: 0

Running Job List

Job Name	Start Time	Duration	End Time	Tasks	Status
SUMO_Madrid_Flink_Traffic	2026-03-10 16:37:28.139	31s 651ms	—	4 / 4	RUNNING



COMPARISON – SPARK VS. FLINK

Rules:

1. Pollution scripts
2. Same arguments: 10min 2min CO2 all
3. Same producer configurations
4. Same runtime - 10min



SPARK

Stats:

```
PS D:\New folder\faks\MASTER\BIGDATA\PROJECT2\docker> docker stats spark-master --no-stream
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
f4fef9f46ca0	spark-master	97.13%	1.257GiB / 7.603GiB	16.53%	19.4MB / 46.4MB	150MB / 125MB	305


```
PS D:\New folder\faks\MASTER\BIGDATA\PROJECT2\docker> docker stats spark-master --no-stream
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
f4fef9f46ca0	spark-master	0.68%	1.242GiB / 7.603GiB	16.34%	19.5MB / 46.7MB	150MB / 125MB	280

- While idle - 0.68% CPU usage
- While processing a mini-batch - 97.13% CPU usage
- Latency: High (waits for batch trigger)
- All the work done by spark-master

```
PS D:\New folder\faks\MASTER\BIGDATA\PROJECT2\docker> docker exec spark-master ps aux | findstr spark
```

root	11	0.4	2.6	2736060	207964	?	Sl	14:49	1:02	/usr/lib/jvm/java-1.8-openjdk/jre/bin/java -cp /spark/conf:/spark/jars/* -Xmx1g org.g.apache.spark.deploy.master.Master --ip f4fef9f46ca0 --port 7077 --webui-port 8080
root	12189	25.5	12.2	4071588	977580	pts/0	Ssl+	18:20	2:34	/usr/lib/jvm/java-1.8-openjdk/jre/bin/java -cp /spark/conf:/spark/jars/* -Xmx2G org.g.apache.spark.deploy.SparkSubmit --packages org.apache.spark:spark-sql-kafka-0-10_2.12:3.1.1 /app/spark_processor.py 10 minutes 2 minutes C02 all
root	12252	0.0	0.2	27888	23712	pts/0	Sl+	18:20	0:00	python3 /app/spark_processor.py 10 minutes 2 minutes C02 all

- Avg. CPU usage - 25.5%
- Avg. memory usage - 12.2%
- Similar results to Flink

SPARK

Note:

```
26/03/10 18:02:55 WARN HDFSBackedStateStoreProvider: The state for version 27 doesn't exist in loadedMaps. Reading snapshot file and delta files if needed...Note that this is normal for the first batch of starting query.  
26/03/10 18:02:55 WARN HDFSBackedStateStoreProvider: The state for version 27 doesn't exist in loadedMaps. Reading snapshot file and delta files if needed...Note that this is normal for the first batch of starting query.  
26/03/10 18:02:55 WARN HDFSBackedStateStoreProvider: The state for version 27 doesn't exist in loadedMaps. Reading snapshot file and delta files if needed...Note that this is normal for the first batch of starting query.  
26/03/10 18:02:55 WARN HDFSBackedStateStoreProvider: The state for version 27 doesn't exist in loadedMaps. Reading snapshot file and delta files if needed...Note that this is normal for the first batch of starting query.  
26/03/10 18:02:55 WARN ProcessingTimeExecutor: Current batch is falling behind. The trigger interval is 10000 milliseconds, but spent 10243 milliseconds
```

The Spark is struggling to keep up with the incoming data rate.

FLINK

Stats:

```
PS D:\New folder\faks\MASTER\BIGDATA\PROJECT2\docker> docker stats flink-jobmanager flink-taskmanager --no-stream
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
da48063a3b79	flink-jobmanager	2.95%	1.391GiB / 7.603GiB	18.30%	24.4MB / 381MB	290MB / 771MB	198
d32624fce8b0	flink-taskmanager	25.71%	1.391GiB / 7.603GiB	18.30%	832MB / 58.4MB	174MB / 395MB	273

Name	Bytes Received	Records Received	Bytes Sent	Records Sent	Parallelism	Start Time	Duration	Tasks
Source: Kafka Emission Source -> Map: _stream_key_by_map_operator	3.46 MB	11,706	4.06 MB	11,702	2	2026-03-10 18:29:11.149	10m 3s 733r	2
SlidingProcessingTimeWindows: Map -> Sink: Writer -> Sink: Committer	4.13 MB	11,702	703 KB	2,465	2	2026-03-10 18:29:11.157	10m 3s 697r	2

- Compression ratio: 11,706 → 2,465 (~79% of data reduction)
- JobManager CPU: **2.95%** | Memory: **1.391 GiB (18.3%)**
- TaskManager CPU: **25.71%** | Memory: **1.391 GiB (18.3%)**
- TaskManager Network I/O: **832 MB in / 58.4 MB out**
- Latency: Low (processes record by record)

THANKS!

A series of several thin, white, parallel diagonal lines extending from the bottom right towards the top right of the slide, adding a modern, geometric touch to the design.